

Project Presentation of CO-OP Project at Industry (Module-2) (22CS421)

On

Detecting Hallucinations in LLMs using Prompt Validation and Response Scoring

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Large Language Models (LLMs) have significantly advanced the field of Artificial Intelligence by generating human-like and context-aware text. They are widely used in applications such as chatbots, virtual assistants, healthcare, education, content generation, and research support systems. Despite their capabilities, LLMs often generate hallucinations, which are responses that are factually incorrect, misleading, or unsupported by reliable information. These hallucinations reduce the reliability and trustworthiness of AI systems, particularly in critical domains where accurate information is essential.

Hallucinations occur because LLMs generate responses based on learned language patterns rather than actual understanding or real-time fact verification. Consequently, models may confidently provide inaccurate or fabricated information, affecting decision-making and user trust. To address this issue, several mitigation approaches such as fine-tuning, response verification, and prompt engineering have been explored. Among these, prompt engineering is considered a simple, scalable, and cost-effective technique for improving response quality.

Therefore, this study focuses on detecting and reducing hallucinations in LLMs using prompt validation and response scoring techniques to improve the reliability, consistency, and trustworthiness of AI-generated content.

Problem Statement

The rapid adoption of Large Language Models (LLMs) in applications such as chatbots, healthcare, education, and research has increased the need for reliable AI-generated content. Although LLMs generate fluent and context-aware responses, they often produce hallucinations, which are factually incorrect, misleading, or unsupported outputs. These hallucinations reduce user trust and limit the safe deployment of LLMs in domains where factual accuracy is critical. Since LLMs rely on learned statistical patterns rather than actual understanding or real-time fact verification, they may confidently generate inaccurate information.

Existing mitigation approaches such as fine-tuning, retrieval-based systems, and external knowledge integration improve factual reliability but often require high computational resources and architectural modifications. Therefore, this research focuses on developing an efficient and scalable approach using Prompt Validation and Response Scoring techniques to detect and reduce hallucinations while improving the reliability, accuracy, and practical usability of LLM-generated responses across real-world applications.

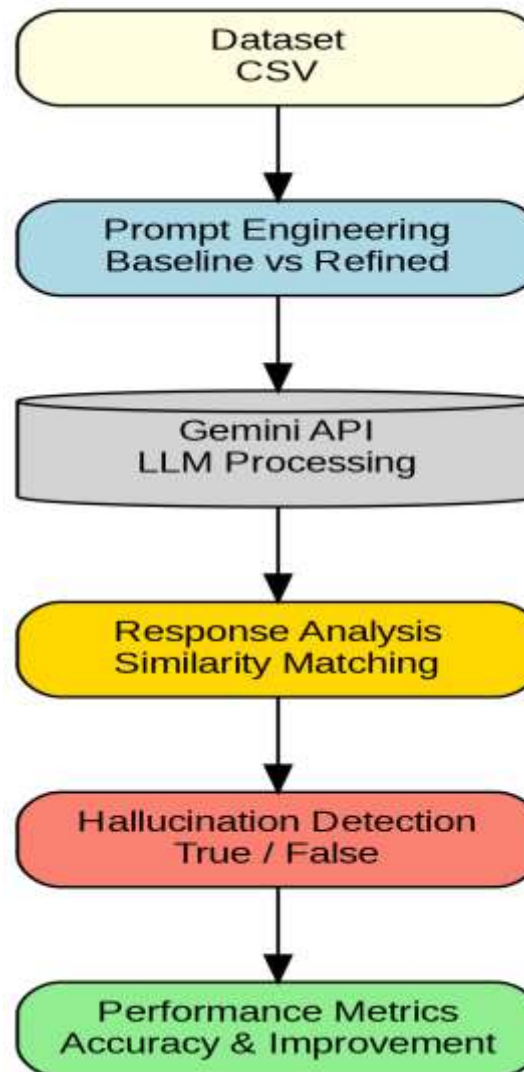
Objectives

- To design and implement a framework for detecting hallucinations in Large Language Models (LLMs) using prompt validation and response scoring techniques
- To analyze the impact of different prompt engineering strategies on the factual accuracy of LLM-generated responses
- To develop structured and context-aware prompts that guide LLMs toward producing more reliable and factually grounded outputs
- To experimentally evaluate hallucination rates across different prompting techniques such as contextual framing, explicit instructions, and structured prompting
- To compare baseline model responses with refined prompt-based responses in order to measure hallucination reduction
- To collect and analyze datasets consisting of factual, reasoning-based, and descriptive prompts for evaluating LLM performance
- To quantify improvements in response reliability using qualitative and quantitative evaluation methods
- To identify common patterns and causes of hallucinations occurring in LLM-generated text across different scenarios
- To optimize prompt engineering strategies that balance response accuracy, coherence, and practical usability
- To propose scalable and cost-effective approaches for improving the trustworthiness and reliability of AI-generated content without requiring model retraining or architectural modifications.

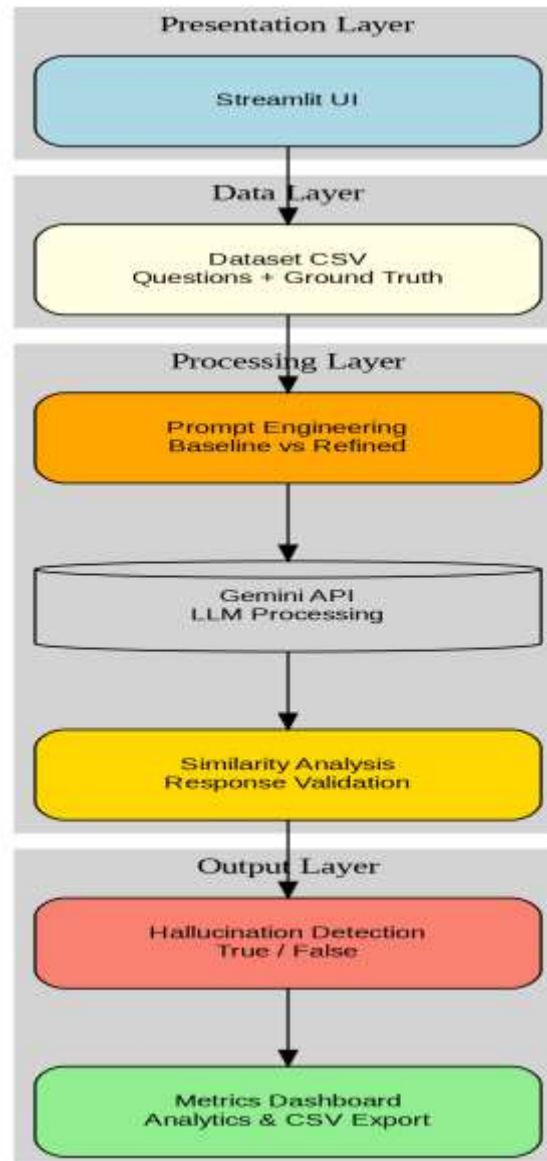
The proposed system is designed as a hallucination detection framework for Large Language Models (LLMs) using prompt validation and response scoring techniques.

- **Prompt Layer**
 - User prompts processed using structured and context-aware prompting
 - Explicit instructions added to reduce ambiguous and hallucinated responses
- **LLM Processing Layer**
 - Responses generated using multiple LLMs such as GPT, Gemini, or Claude
 - Baseline and refined prompts compared for evaluation
- **Validation Layer**
 - Responses checked using scoring and validation mechanisms
 - Hallucinated or unsupported outputs identified and analyzed
- **Monitoring & Analysis**
 - Hallucination rates, response accuracy, and consistency recorded
 - Results visualized using graphs and comparative analysis
- **Deployment Model**
 - API-based scalable framework supporting multiple LLM providers
 - Suitable for chatbot, research, and AI-assisted applications

System Architecture



High Level Design (HLD)



Implementation Strategy

- **Prompt Engineering Framework**
 - Designed a hallucination detection framework using prompt validation and response scoring techniques
 - Implemented contextual prompting, structured prompts, and explicit instruction-based inputs
 - Used multiple LLM APIs for comparative response generation and evaluation
- **Experimental Evaluation Approach**
 - Compared baseline prompting vs refined prompt-engineered models
 - Conducted controlled experiments to measure hallucination reduction and response reliability
 - Simulated factual, reasoning, and descriptive query scenarios
- **Performance Metrics Evaluated**
- **Hallucination Rate:** Incorrect or unsupported responses
- **Response Accuracy:** Factual correctness of outputs
- **Response Consistency:** Stability of generated responses
- **Prompt Effectiveness:** Impact of prompt refinement strategies
- **Validation & Reliability Measures**
 - Implemented response scoring and verification mechanisms
 - Used qualitative and quantitative evaluation techniques
 - Applied statistical analysis and comparative visualization for result interpretation
- **Dataset & Analysis**
 - Generated a structured prompt dataset covering multiple query categories
 - Responses collected and processed using Python-based evaluation pipelines
 - Data analyzed using graphs, tables, and comparative summaries to identify hallucination patterns

Methodology

- **Prompt Dataset Generation**
 - Created a dataset of prompts covering factual, reasoning, and descriptive tasks
 - Designed prompts to simulate real-world interactions with LLMs
 - Responses collected from multiple models under different prompting conditions
- **Scenario Design**
- **Baseline Model:** Standard prompts without refinement
- **Refined Prompt Model:** Prompt-engineered inputs using contextual framing, structured prompts, and explicit instructions
- Each prompt evaluated under both scenarios for comparison
- **Evaluation Parameters**
- **Hallucination Rate (%):** Incorrect or unsupported responses
- **Response Accuracy:** Factual correctness of outputs
- **Response Consistency:** Stability and reliability of responses
- **Prompt Effectiveness:** Impact of prompt engineering on reducing hallucinations
- **Experimental Process**
 - Prompts executed using automated scripts and LLM APIs
 - Responses collected, validated, and scored for hallucinations
 - Results recorded for analysis and comparison
- **Comparative Analysis**
 - Compared baseline and refined prompt responses
 - Performed quantitative and qualitative analysis of hallucination reduction
 - Visualized results using graphs and summary tables

Advantages and Disadvantages

- **Advantages (Prompt Validation & Hallucination Detection Framework)**

- Reduces hallucination rates in LLM-generated responses significantly
- Improves factual accuracy and reliability of AI-generated content
- Enhances trustworthiness of LLMs in sensitive applications such as healthcare, education, and research
- Provides a cost-effective solution without requiring model retraining or architectural modifications
- Encourages context-aware and structured response generation through prompt engineering
- Improves response consistency by reducing speculative and misleading outputs
- Enables better validation and scoring of generated responses for detecting unreliable information
- Supports scalable deployment across different LLMs and application domains
- Enhances user trust and confidence in AI-assisted systems
- Promotes the development of safer, more responsible, and ethically reliable AI systems

- **Disadvantages / Trade-offs**

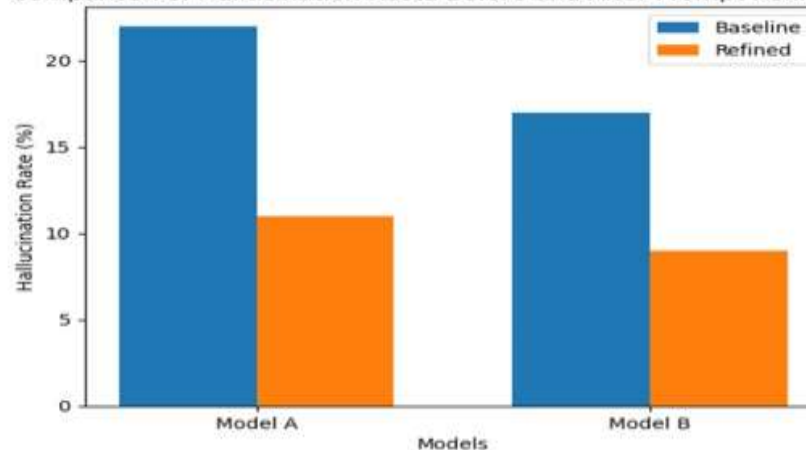
- Prompt engineering effectiveness depends heavily on prompt quality and design expertise
- Hallucinations cannot be completely eliminated, especially in complex or ambiguous scenarios
- Different LLMs may respond inconsistently to the same prompt structure
- Increased prompt complexity can lead to longer response generation times
- Requires continuous prompt refinement and evaluation for maintaining accuracy
- Response scoring and validation mechanisms may introduce additional computational overhead
- Limited effectiveness in handling highly domain-specific or newly emerging information
- Manual evaluation of hallucinations can be time-consuming and subjective
- Over-constrained prompts may reduce creativity and flexibility in generated responses
- Reliability still depends on the underlying training data and limitations of the LLM itself

Conclusion



- This work establishes a structured framework for detecting and reducing hallucinations in Large Language Models (LLMs) using prompt validation and response scoring techniques. The proposed approach improves response reliability by embedding contextual guidance, structured prompting, and explicit verification mechanisms directly into the interaction process, rather than relying solely on model retraining or architectural modifications.
- Experimental evaluation conducted across multiple prompt categories, including factual recall, reasoning, and descriptive tasks, demonstrates that prompt-engineering-based interventions significantly reduce hallucination rates in LLM-generated responses. The framework achieves nearly a 50% reduction in hallucinations, with Model A improving from approximately 22% to 11% and Model B from 17% to 9%. These improvements are primarily attributed to the use of structured prompts, contextual framing, and explicit instructions that constrain speculative response generation and encourage fact-aware outputs.
- Overall, the findings validate that prompt engineering and response validation provide a scalable, cost-effective, and practical approach for mitigating hallucinations in LLMs. The proposed framework demonstrates that reliability and factual correctness can be significantly enhanced without additional model training, positioning prompt-driven mitigation strategies as an effective solution for deploying trustworthy AI systems in real-world applications.

Comparison of Hallucination Rates Before and After Prompt Refinement



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1003	Detecting Hallucinations in LLMs using Prompt Validation and Response Scoring View abstract Currently, large language models (LLMs) can generate contextually relevant fluent text. Nonetheless, they tend to 'hallucinate' by providing answers that contain false or unverifiable information. This research evaluates the utility of various prompt engineering techniques for mitigating hallucinated outputs and enhancing LLM generated content accuracy and reliability. A total of 100 prompts were created and analyzed for LLM responses to fact-based queries, those requiring logical reasoning, and content generation prompts of a descriptive nature. In the first step, two of the most popular LLMs were used to assess their response to basic prompts. Responses to the iteratively improved prompts, which included explicit instructions, constraints on the responses, verification and uncertainty management steps, were compared to responses to the basic prompts. It was noted that hallucinations decreased with each prompt improvement step. LLM A's hallucination rate improved from 22% to 11%, while LLM B's from 17% to 9%, resulting in an average improvement of close to 50%. Prompts that instructed the LLM to perform a self-check and recognize uncertainty generated more reliable responses than traditional prompt techniques. Overall, these results indicate that prompt design can effectively improve LLMs reliability, and more so, that such improvements can be leveraged at a low cost to develop reliable AI systems.	Track 2: Artificial Intelligence, Data Science & Cybersecurity Governance	Submission files: ④ Detecting Hallucinations in LLMs using prompt validation and response scoring (1).pdf	Submission: ✎ Edit 📄 Submission 🔍 Edit Conflicts

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Track Name: Track 2: Artificial Intelligence, Data Science & Cybersecurity Governance

Paper ID: 1003

Paper Title: Detecting Hallucinations in LLMs using Prompt Validation and Response Scoring

Abstract:

Currently, large language models (LLMs) can generate contextually relevant fluent text. Nonetheless, they tend to 'hallucinate' by providing answers that contain false or unverifiable information. This research evaluates the utility of various prompt engineering techniques for mitigating hallucinated outputs and enhancing LLM generated content accuracy and reliability. A total of 100 prompts were created and analyzed for LLM responses to fact-based queries, those requiring logical reasoning, and content generation prompts of a descriptive nature. In the first step, two of the most popular LLMs were used to assess their response to basic prompts. Responses to the iteratively improved prompts, which included explicit instructions, constraints on the responses, verification and uncertainty management steps, were compared to responses to the basic prompts. It was noted that hallucinations decreased with each prompt improvement step. LLM A's hallucination rate improved from 22% to 11%, while LLM B's from 17% to 9%, resulting in an average improvement of close to 50%. Prompts that instructed the LLM to perform a self-check and recognize uncertainty generated more reliable responses than traditional prompt techniques. Overall, these results indicate that prompt design can effectively improve LLMs reliability, and more so, that such improvements can be leveraged at a low cost to develop reliable AI systems.

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